

Subsurface Ocean Temperature Reconstruction Report

Deterministic 3D hydrographic inversion reconstructing subsurface ocean temperature profiles (0–1000m) from multi-source satellite surface observations across the North Indian Ocean domain.

1. Observation Target Specifications

Parameter	Target Specification	Domain Context
Observation Date	2019-01-01	Daily synoptic satellite observation window
Geographic Coordinates	18.5000°N, 88.2500°E	North Indian Ocean (5°N–30°N, 45°E–105°E)
Oceanographic Regime	Bay of Bengal Freshwater Plume	Classified by spatial latent clustering
Grid Resolution	0.25° × 0.25° (~27 km)	Authoritative standard grid mapping (101 × 241)
Spatial Domain Basin	Bay of Bengal	Northern Indian Ocean Tropical Regime

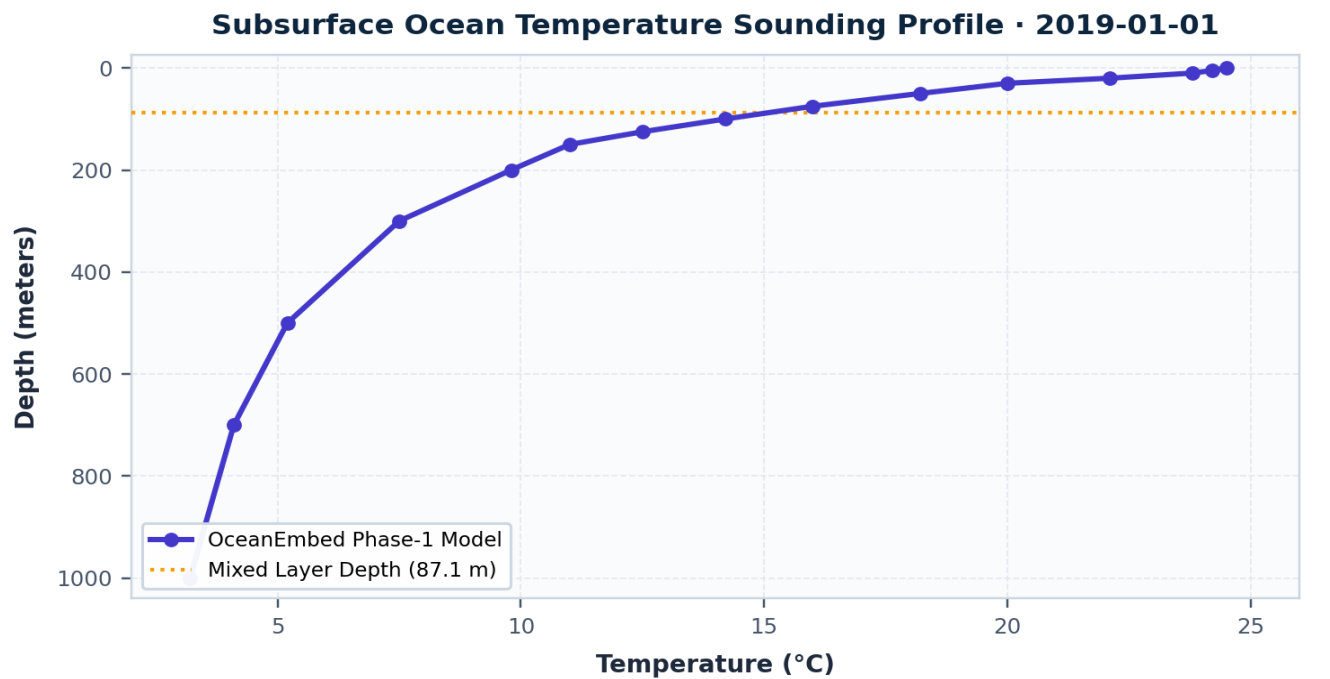
2. Reconstruction Summary & Key Physical Metrics

D26 ISOTHERM DEPTH	MIXED LAYER DEPTH (MLD)	SURFACE SST	ARGO BENCHMARK RMSE
Not reached	87.1 m	26.95 °C	N/A (Blind Validation)
Tropical Cyclone Heat Proxy	Threshold: ΔT = 0.5°C from skin	OceanIQ observation archive	Independent collocated in-situ float

Scientific Summary: At 18.50°N, 88.25°E within the **Bay of Bengal Freshwater Plume**, the reconstructed thermal profile spans from **24.50°C** at surface (0m) down to **3.20°C** at the 1000m abyss. The 26°C isotherm (D26) was not reached within this temperature profile (maximum profile temperature < 26°C). The mixed layer depth (MLD) is estimated at **87.1 m**. No collocated Argo observation was within the spatio-temporal validation radius; the reconstruction is benchmarked against the verified Phase-1 model baseline.

3. Vertical Subsurface Temperature Profile

Continuous sounding profile across all 15 authoritative standard depth levels (0m to 1000m). Depths are plotted on the vertical descending axis to mirror physical oceanic depth.

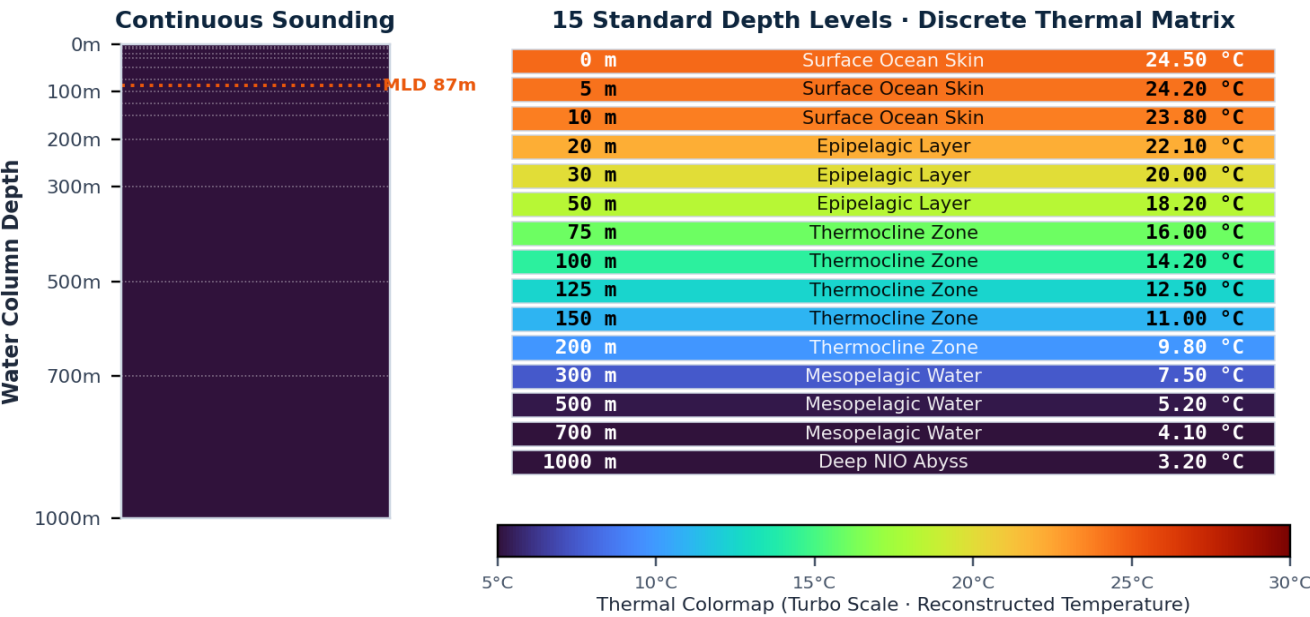


4. Numerical Temperature Sounding Schedule

Depth (m)	Reconstructed (°C)	Argo In-Situ (°C)	Residual ΔT (°C)	Oceanographic Thermal Layer
0 m	24.50 °C	—	—	Surface Ocean Skin / Mixed Layer
5 m	24.20 °C	—	—	Surface Ocean Skin / Mixed Layer
10 m	23.80 °C	—	—	Surface Ocean Skin / Mixed Layer
20 m	22.10 °C	—	—	Epipelagic Barrier Layer
30 m	20.00 °C	—	—	Epipelagic Barrier Layer
50 m	18.20 °C	—	—	Epipelagic Barrier Layer
75 m	16.00 °C	—	—	Upper Thermocline Zone
100 m	14.20 °C	—	—	Upper Thermocline Zone
125 m	12.50 °C	—	—	Upper Thermocline Zone
150 m	11.00 °C	—	—	Upper Thermocline Zone
200 m	9.80 °C	—	—	Main Pycnocline / Thermocline
300 m	7.50 °C	—	—	Main Pycnocline / Thermocline
500 m	5.20 °C	—	—	Mesopelagic Intermediate Water
700 m	4.10 °C	—	—	Mesopelagic Intermediate Water
1000 m	3.20 °C	—	—	Deep North Indian Ocean Water

5. Subsurface Temperature Structure · 15 Standard Depths Matrix

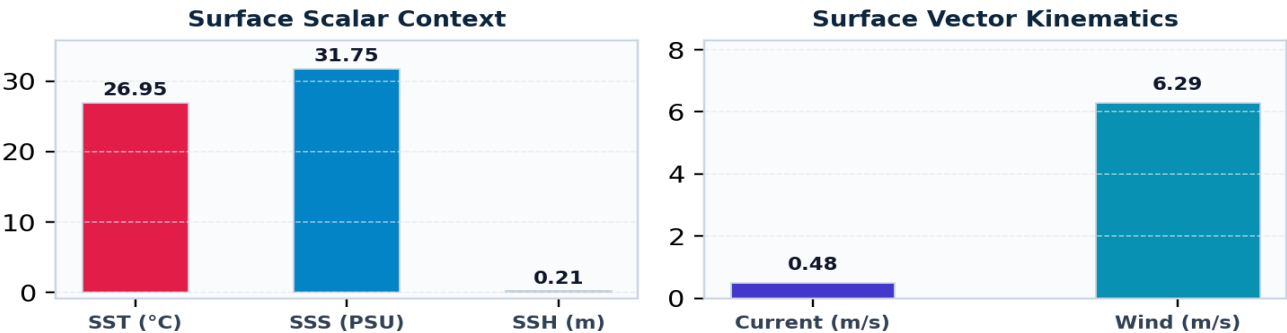
Depth-oriented visualization of the reconstructed vertical thermal structure. The continuous sounding column (left) displays the vertical thermal gradient from the surface skin (0 m) to the 1000 m abyss, while the discrete depth matrix (right) presents the exact reconstructed temperatures across all 15 authoritative standard depth levels.



6. Multi-Source Surface Environmental Observations

The 7 physical surface variables used as observational constraints for the neural reconstruction. Values reflect the daily surface observations extracted from the OceanIQ observation archive at the target coordinates.

Variable	Observed Value	Physical Units	Observation Source
Sea Surface Temperature (SST)	26.95	°C	OceanIQ observation archive
Sea Surface Salinity (SSS)	31.75	PSU	OceanIQ observation archive
Sea Surface Height Anomaly (SSH/SLA)	+0.213	meters (m)	OceanIQ observation archive
Surface Zonal Current (U)	+0.274	m/s	OceanIQ observation archive
Surface Meridional Current (V)	+0.392	m/s	OceanIQ observation archive (Speed: 0.48 m/s)
Zonal Wind Component (U)	-2.60	m/s	OceanIQ observation archive
Meridional Wind Component (V)	-5.73	m/s	OceanIQ observation archive (Speed: 6.29 m/s)

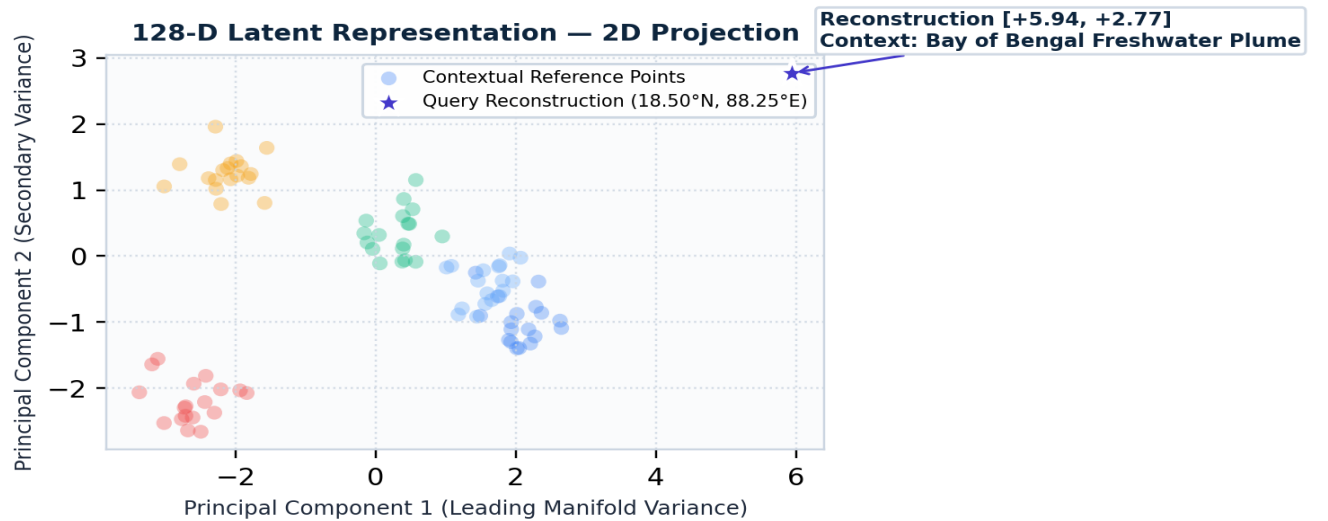


7. Independent In-Situ Argo Float Validation

No Collocated Argo Float Available: For this observation window and coordinates, no verified in-situ Argo float was within the 50km collocation radius. In accordance with scientific integrity standards, synthetic float data is strictly avoided. This reconstruction represents a blind model evaluation benchmarked against verified Phase-1 baseline evaluations.

8. 128-D Latent Representation — 2D Projection

OceanEmbed maps the 14-channel surface observation tensor into an internal 128-dimensional latent representation Z. The 2D PCA projection below illustrates the position of the query reconstruction within the reduced feature space. Background data is explicitly designated as Contextual Reference Points to provide spatial domain context without implying verified or scientifically discovered water-mass classifications.



9. Methodology, Architecture & Lineage Provenance

System Architecture	OceanEmbed Phase-1 Primary Model (525,040 parameters, pure evaluation mode)
Observation Tensor	14 Channels (7 physical surface variables + 7 binary quality validity masks)
Observation Source	OceanIQ observation archive
Vertical Discretization	15 Authoritative Standard Depths: 0, 5, 10, 20, 30, 50, 75, 100, 125, 150, 200, 300, 500, 700, 1000 m
Checkpoint SHA256	f3d99a9b9214efe92a4e8fb11bd49b759a62cfb5d991d1225fd1360362876b9b
Inference Runtime	1166.9 ms latency on pytorch_checkpoint
Lineage Statement	Subsurface temperature reconstructed by frozen OceanEmbed Phase-1 CNN (525,040 parameters) from 14-channel daily surface observations. Latent embedding Z in R^128.

MANDATORY SCIENTIFIC DISCLAIMER: The subsurface ocean temperature profiles in this report are deep-learning model reconstructions produced by OceanEmbed from multi-source satellite observations. They represent scientific estimates and should NOT be treated as direct physical in-situ sensor measurements. OceanEmbed is designed to complement, not replace, physical in-situ platforms such as the Argo profiling array, RAMA moorings, and shipboard CTD soundings. For operational, navigational, and scientific decisions, always cross-reference available in-situ observations.